

June 24, 2026

To the Editor-in-Chief and the Editorial Board,
IEEE Transactions on Network and Service Management

Subject: Submission of an original manuscript for consideration as a regular paper.

Dear Editor,

We are pleased to submit our manuscript, “**Learning to Route Under Load: An Inductive Graph Network that Amortizes Congestion-Aware Traffic Engineering for LEO Mega-Constellations**,” for consideration for publication in the *IEEE Transactions on Network and Service Management*.

Problem. Low Earth orbit (LEO) mega-constellations route traffic over a large, fast-changing mesh of inter-satellite links whose topology is known in advance from ephemeris. Under load, routing on propagation delay alone overloads a few bottleneck links while alternatives sit idle, inflating end-to-end travel time by up to a factor of forty in our study. The congestion-aware optimum, the user equilibrium (UE), removes this waste but requires the full demand matrix and an iterative solve every slot, which does not scale to constellations of more than a thousand satellites with seconds-scale topology change. This is a network-management problem: how to obtain congestion-aware routing at constellation scale without paying for a per-slot optimization.

Contribution. We formalize routing under load as the *amortization* of a traffic-assignment equilibrium and solve it with an inductive graph neural network that predicts the equilibrium congestion price on every link in a single forward pass from cheap, observable load features, decoded by a one-shot multipath router. The method:

- closely approaches UE in distribution, recovering about 97% of the gain over blind routing;
- transfers *zero shot* to a shell twelve times larger, still recovering $94 \pm 2\%$ of the gain, remaining the best deployable policy;
- runs 21 to $29\times$ faster than the iterative solve, with the saving *growing* with constellation size;
- sustains low travel time under drifting traffic hotspots when fed predicted demand, where a reactive controller degrades below congestion-blind routing.

We compare against not only the natural cheap heuristics (blind, geographic, one-step) and the UE/system-optimum references, but also a *learned* alternative that predicts the routing directly; our price-field approach is both more accurate and one to two orders of magnitude cheaper at inference. We further contribute (i) a proposition bounding the travel-time suboptimality of the decoded routing by the price-prediction error, validated empirically; and (ii) a documented set of evaluation pitfalls specific to learned traffic engineering, with a protocol to avoid them. We also report, transparently, a negative result: a quantum-walk propagation operator that motivated the study offers no benefit once load features are available.

Significance and fit. To our knowledge this is the first method that amortizes congestion-aware traffic engineering for LEO at mega-constellation scale with a verified, inductive, zero-shot guarantee of quality, and the inductive transfer to a $12\times$ larger shell is, we believe, a result of independent interest. The work lies squarely within the scope of *IEEE TNSM*: congestion-aware traffic engineering, learning-based network control, and resource management for non-terrestrial networks. It

pairs a management-oriented formulation with a deployable, inexpensive control mechanism and an honest, reproducible evaluation methodology.

Reproducibility. The complete simulator, models, experiments, and figure-generation scripts are publicly released; every reported number is regenerated from result files by an included script, and all results follow from constellation parameters with no external data dependency.

Declarations. This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere, in whole or in part. All authors have read and approved the submission and agree to its content. The authors declare no conflict of interest. This research is funded by the Science and Technology Development Fund of The University of Danang under project number B2025-DN02-26.

Thank you for considering our submission. We look forward to the reviewers' feedback.

Sincerely,

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